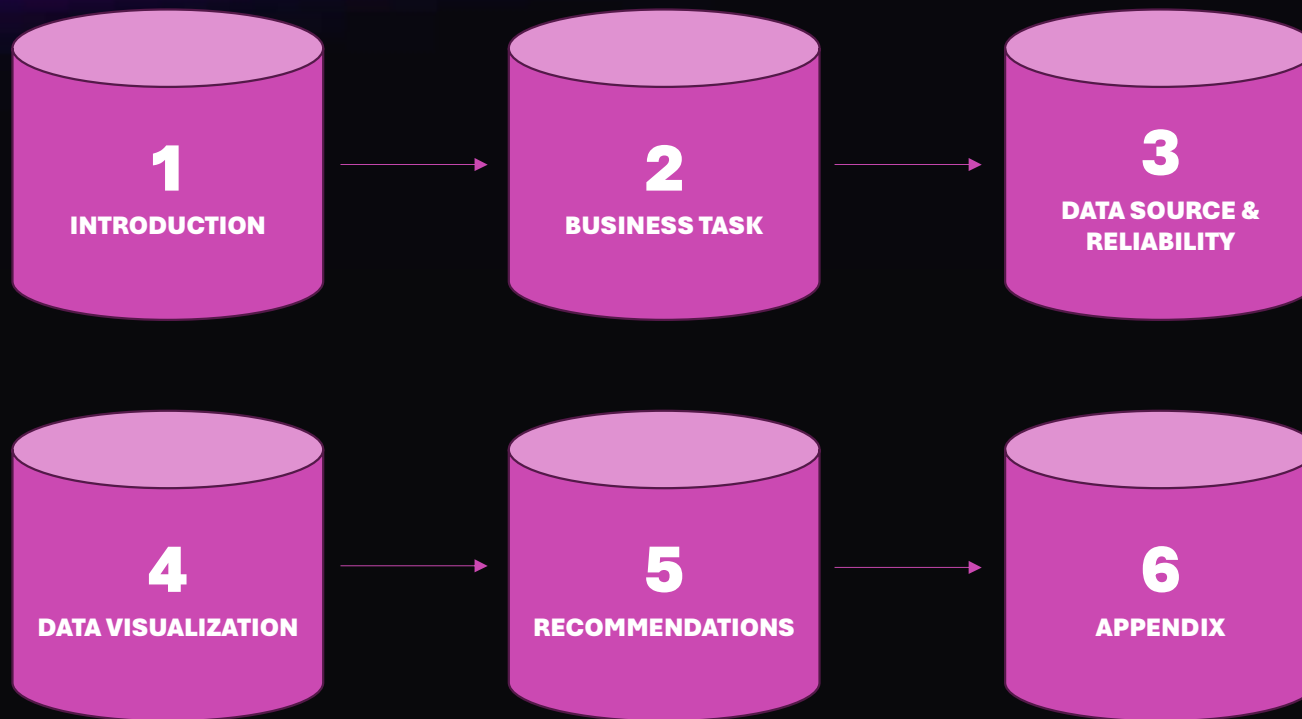


Bellabeat

Case Study: Analyzing Daily Consumer Fitness Habits

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INTRODUCTION

Bellabeat is a pioneering wellness company that designs high-tech, health-focused smart products for women. To support the company's ongoing growth, this project analyzes historical fitness tracker data to uncover the real-world daily habits of smart device users.

By examining daily step volumes, calorie burn rates and weekly activity routines, this analysis bridges the gap between raw numbers and actual consumer behavior, providing a clear picture of how and when users truly engage with their devices.

BUSINESS TASK

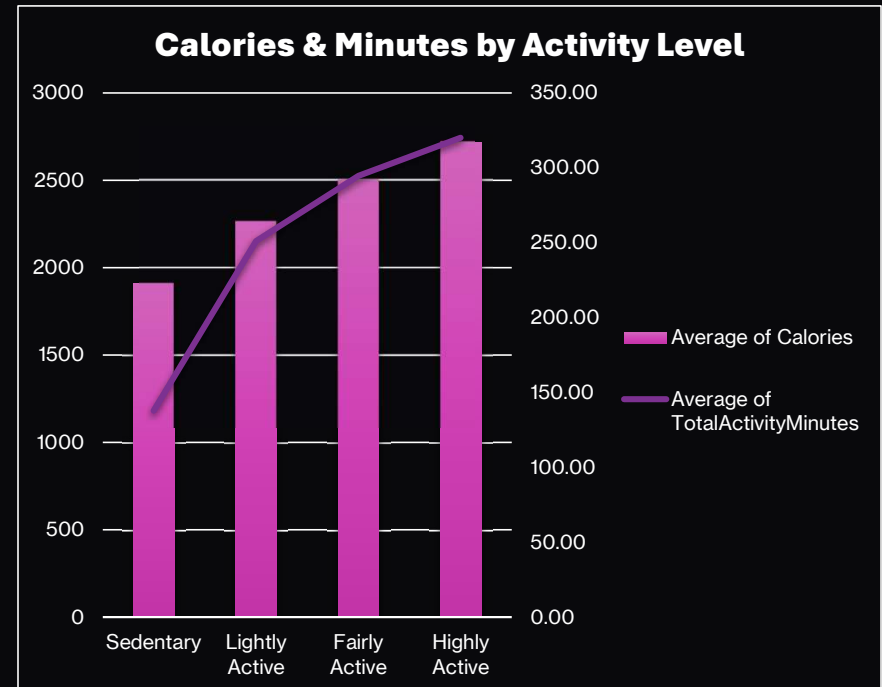
- **The Task:** Analyze smart fitness device data to uncover core consumer habits, activity levels and daily usage trends.
- **Objective:** Translate these data insights into targeted, actionable marketing strategies that boost user engagement, encourage daily consistency and drive Bellabeat's future growth.

Data Source & Reliability

- **Origin:** Public FitBit Fitness Tracker dataset (via Kaggle).
- **Link:** <https://www.kaggle.com/datasets/arashnic/fitbit>
- **Volume:** 1.165 logged days across ~ 35 unique users.
- **Key Metrics:** Device-recorded daily steps, active minutes and calories burned.
- **Integrity:** Device-generated data eliminates human-reporting bias.
- **Privacy:** All users IDs are completely anonymized.
- **Limitations:** Small sample size (~ 35 users) and no demographic data available (e.g. age, gender).

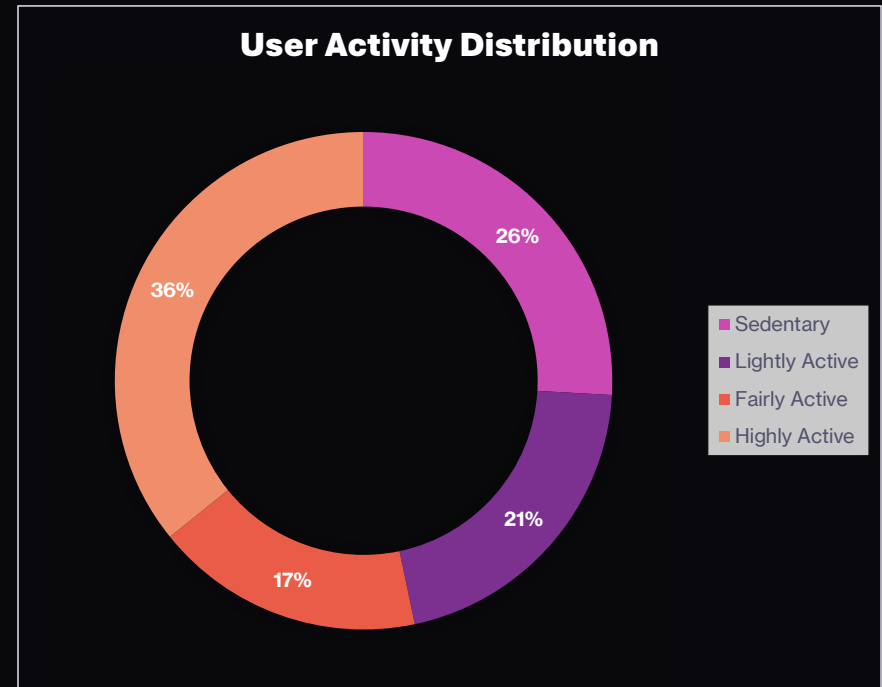
Calories & Minutes by Activity Level

- **Key Takeaway:** Moving more directly leads to burning more calories, with a big jump when users reach the highest activity level.
- Sedentary (inactive) days average about 1,900 calories burned and under 150 active minutes.
- Highly Active days average over 2,700 calories burned and more than 300 active minutes.



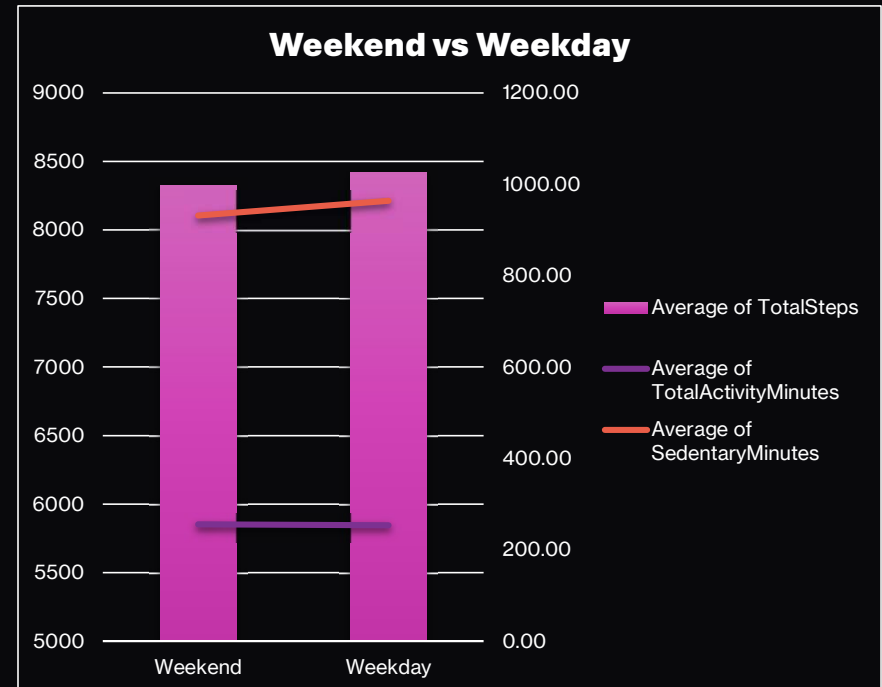
User Activity Distribution

- **Key Takeaway:** Users are mostly either very active or not active at all, with very few “average” days in between.
- “Highly Active” is the most common daily level, making up 36% of all tracked days.
- “Sedentary” (inactive) days are the second most common at 26%.
- The middle levels are much lower, with “Lightly Active” at 21% and “Fairly Active” at 17%.



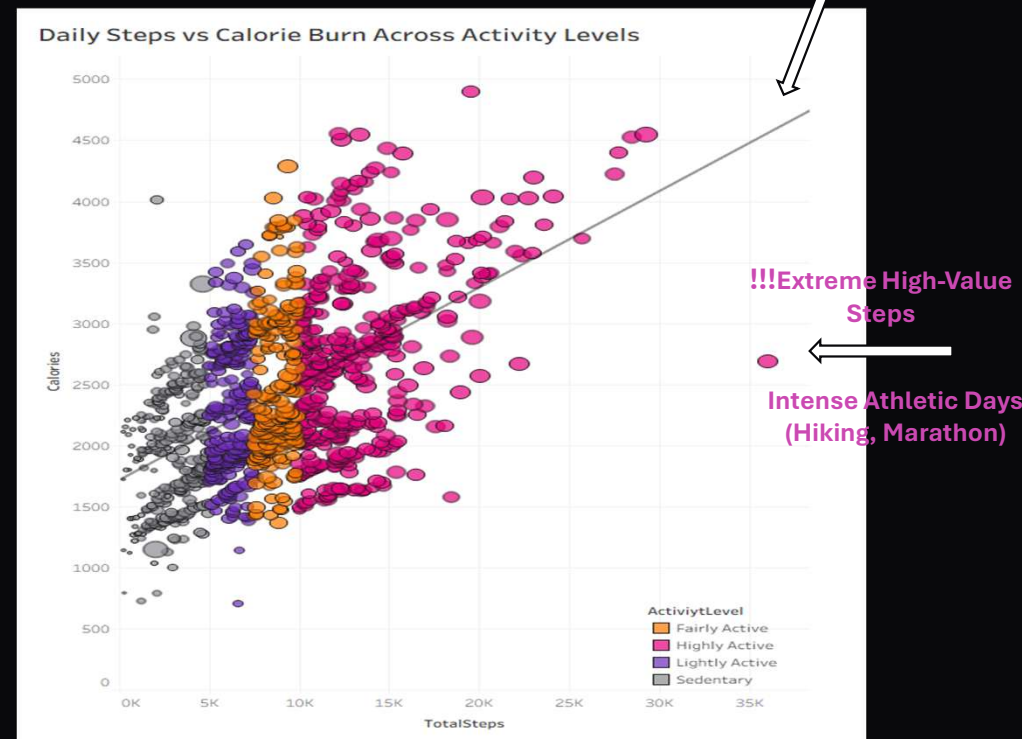
Weekend vs Weekday

- **Key Takeaway:** Users have the same activity habits every day, proving they do not just save their workouts for the weekends.
- Average step counts stay very steady, with users taking about 8,400 steps on weekdays and 8,300 on weekends.
- Total active time is almost the same, staying right around 250 minutes no matter the day.



Correlation Between Steps & Calories

- **Key Takeaway:** Hitting the 10,000 steps mark is the best way to see a big boost in total calories burned.
- The chart shows a clear link: taking more steps and staying active longer burns more calories.
- Reaching the “Highly Active” level (over 10,000 steps) reliably pushes burned calories well above 2,500 for the day.



Weekly Activity Level Heatmap

- **Key Takeaway:** Users are most active in the middle of the week but lose motivation on Fridays and Sundays.
- “Highly Active” days happen most often mid-week, peaking at 66 days on Wednesdays.
- Completely “Sedentary” (inactive) days spike at the end of the week, reaching 51 days on Fridays and high of 57 days on Sundays.

Weekly Activity Level Heatmap

		Day_Of_Week						
		Monday	Tuesday	Wednes..	Thursday	Friday	Saturday	Sunday
ActivityLevel	Sedentary	36	32	41	38	51	47	57
	Lightly Active	35	35	38	27	36	36	35
	Fairly Active	28	27	30	39	39	20	21
	Highly Active	62	60	66	59	53	66	51

Recommendations

- **Send Reminders on Fridays and Sundays:** Users are most inactive on these specific days, with completely sedentary days peaking at 51 on Fridays and 57 on Sundays. Bellabeat should send notifications or mini-challenges to motivate users when they usually skip their workouts.
- **Reward Hitting 10,000 Steps:** Reaching the 10,000-step mark is the proven tipping point for burning over 2,500 calories a day. The Bellabeat app should use milestone badges and rewards to encourage average users to push just a little harder and cross this daily goal.
- **Focus on Everyday Habits:** Users do not save their workouts for the weekend. They average about 8,400 steps on weekdays and 8,300 on weekends. Marketing campaigns should stop focusing on intense “weekend warrior” workouts and instead promote building consistent, healthy daily routines.

Appendix

Data Aggregation & Formatting

- Merged multiple raw activity files into a single spreadsheet and standardized all dates into a DD/MM/YY format for consistency.

Data Cleaning

- Removing 24 duplicate log entries.
- Filtering out days where the device was clearly not worn, such as days with exactly 1,440 sedentary minutes or less than 600 total wear minutes.
- Removing incomplete tracking days, such as days showing 0 steps, less than 200 total steps, or under 200 calories burned (which indicates partial tracking or a sync failure).
- Legitimate extreme high-value days (e.g., marathons, hiking days with over 20,000 steps) were kept in the dataset.

Data Engineering

- Created new custom variables to analyze user habits better.
- Calculated total wear minutes and total active minutes.
- Extracted the specific “Day of Week” and added an “Is_Weekend” identifier to compare weekday vs. weekend behaviors.
- Built a custom “Activity Level” category to divide users into four distinct tiers based on their daily step counts: Sedentary, Lightly Active, Fairly Active and Highly Active.

THANK YOU